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Invitation to Review for IEEE Access Access-2024-22721

1 message

IEEE Access <onbehalf@manuscriptcentral.com>

Sat, Jul 27, 2024 at 12:21 AM

Reply-To: ieeeaccess2@ieee.org

To: khare.pranav@gmail.com

27-Jul-2024

Dear Mr. Khare:

The manuscript entitled "Beyond Classical Cryptography: A Systematic Review of Post-Quantum Hash-Based Signature Schemes, Security, and Optimizations" has been submitted to IEEE Access. Since your expertise is relevant to this article, we would be grateful if you would consider reviewing it.

The abstract appears at the end of this letter, along with the names of the authors. Please let me know within 2 days if you will be able to accept my invitation to review. If you agree to review, we ask that you provide your review comments within **7 days** to assist IEEE Access in maintaining its expedited peer review process of 4 to 6 weeks. If you are interested in reviewing but need additional time to complete the review, we can grant a reasonable deadline extension upon request. For additional information on IEEE Access and tips on completing a quality review, please visit our [Reviewer Best Practices guide](#).

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Once you accept my invitation to review this manuscript, you will be notified via email with directions on how to access the manuscript and the reviewer instructions in the Reviewer Center.

Thank you for your assistance with the peer review of this IEEE Access article.

Sincerely,

Prof. Derek Abbott
Editor-in-Chief, IEEE Access

ieeeaccess2@ieee.org

MANUSCRIPT DETAILS

TYPE: Survey

TITLE: Beyond Classical Cryptography: A Systematic Review of Post-Quantum Hash-Based Signature Schemes, Security, and Optimizations

AUTHORS: Fathalla, Efat; Azab, Mohamed

ABSTRACT: The emergence of quantum computing poses significant risks to the security of current cryptographic systems, particularly those reliant on classical algorithms vulnerable to quantum attacks. This systematic literature review adopts the PRISMA model to critically assess the development, methodologies, and security of post-quantum hash-based signature schemes as resilient alternatives. Through a methodical selection process from leading academic databases, we identify and analyze key contributions to the field within the last decade, focusing on the schemes' security proofs, enhanced performance, and efficiency metrics. Our analysis reveals a diverse landscape of hash-based signature schemes, their evolving security features against quantum threats, and their practical implementations in securing digital communications. The review highlights the importance of advancing these quantum-resistant technologies, discusses the challenges in their adoption, and outlines future directions for research and standardization efforts. The findings aim to provide a comprehensive resource for researchers, practitioners, and policymakers involved in the transition toward secure cryptographic practices in the quantum era.

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If you have any questions, please contact the article administrator: Dr. Manpreet Kaur m.kaur@ieee.org. In addition, if you would like to get a certificate after completing this review, please send an email to ieeeaccess@ieee.org.

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